

# Recursive Entrainment and Latent Synchronization Regimes in Autoregressive Language Models

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## Abstract

Autoregressive language models exhibit measurable latent synchronization regimes characterized by architecture- and training-dependent recursive entrainment thresholds, frequency-selective resonance spectra, and attractor dynamics. We introduce **MASSIF** (Multiscale Attractor Stability & Stress Inference Framework), a telemetry system for Computational Latent Dynamics - the study of emergent dynamical behavior in high-dimensional learned systems.

Across 8 architectures (Bloom-560M, GPT-2 family, TinyLlama-1.1B, DeepSeek-1.3B, Gemma-2-2B), we perform high-statistics sweeps ( $N \geq 30$  per condition) across temperature, block resonance, Fourier spectra, and noise response (pink, white, chirp).

We identify three distinct dynamical families: **broadband resonators** (GPT-2 family: high pink noise sensitivity > 75%), **narrowband resonators** (Bloom, TinyLlama: low pink noise, high chirp > 45%), and **high-frequency selective** (DeepSeek: minimal response). GPT-2 Medium (355M) is anomalous within its family (7% flip vs 53-80% for others), suggesting a possible critical scale.

We propose **recursive entrainment** as the unifying concept: persistent latent directional coherence induced by phase-alignment between hidden-state trajectories and periodic or self-referential prompt structure. Results suggest that recursive susceptibility behaves spectrally - models respond to dynamical structure (periodicity, recurrence, entropy spectra, synchronization pressure) rather than semantic content alone. These findings have implications for AI safety, long-context stability, and the design of robust generative systems.

## 1 Introduction

Autoregressive language models operate as high-dimensional dynamical systems. Under ordinary prompting, hidden-state evolution remains stable. However, recursive or self-referential inputs can induce pathological attractor behavior: repetition collapse, degeneration loops, mode-locking, and unstable latent trajectories.

We introduce **MASSIF** (Multiscale Attractor Stability & Stress Inference Framework), a telemetry system within **Computational Latent Dynamics** - the study of emergent dynamical behavior, attractor structure, stability regimes, and recursive phase phenomena in high-dimensional learned systems.

Our central finding is that recursive susceptibility behaves spectrally, not symbolically. Models respond to dynamical structure (periodicity, recurrence, entropy spectra, synchronization pressure) rather than semantic content alone.

## 2 The MASSIF Framework

### 2.1 Hidden-State Dynamics

Let  $h_t \in \mathbb{R}^d$  be the hidden state at step  $t$ :

$$\hat{h}_t = \frac{h_t}{\|h_t\| + \varepsilon}, \quad v_t = \hat{h}_t - \hat{h}_{t-1}$$

## 2.2 Persistence $I_t$ and Flip Criterion

Persistence measures directional continuity:

$$I_t = \frac{v_t \cdot v_{t-1}}{\|v_t\| \|v_{t-1}\| + \varepsilon}$$

$I_t < 0$  indicates corrective anti-persistence;  $I_t > 0$  indicates attractor persistence. A **persistence flip** is defined as  $\text{mean}(I_{t-3:t}) > 0$ .

## 2.3 Derived Quantities

$$\sigma_t = \text{std}(\{I_{t-k:t}^{(r)}\}_{r=1}^R), \quad \Phi_t = \sigma_t - \sigma_{t-1}, \quad \Psi_t^{(m)} = \frac{1}{m} \sum_{i=0}^{m-1} \Phi_{t-i}, \quad \kappa_t = \arccos\left(\frac{v_t \cdot v_{t-1}}{\|v_t\| \|v_{t-1}\|}\right)$$

## 3 Recursive Entrainment: Unifying Concept

We define **recursive entrainment** as the phenomenon where autoregressive hidden-state trajectories become phase-aligned with externally imposed periodic or self-referential prompt structure, producing persistent latent directional coherence and attractor lock-in.

Table 1: Recursive entrainment framework

| Perturbation Type  | Entrainment Mechanism |
|--------------------|-----------------------|
| Lexical repetition | Periodic forcing      |
| Semantic recursion | Recursive forcing     |
| Pink noise         | Scale-free forcing    |
| Chirp              | Frequency sweep       |
| Temperature        | Entropy injection     |

A persistence flip ( $I_t$  crossing from negative to positive) corresponds to the entrainment threshold crossing.

## 4 Experimental Design

### 4.1 Models Evaluated

Table 2: Models evaluated

| Model          | Architecture | Size |
|----------------|--------------|------|
| Bloom-560M     | Transformer  | 560M |
| GPT-2 Small    | Transformer  | 124M |
| GPT-2 Medium   | Transformer  | 355M |
| GPT-2 Large    | Transformer  | 774M |
| GPT-2 XL       | Transformer  | 1.5B |
| TinyLlama-1.1B | Transformer  | 1.1B |
| DeepSeek-1.3B  | Transformer  | 1.3B |
| Gemma-2-2B     | Transformer  | 2B   |

### 4.2 Statistical Protocol

- $N = 30$  runs per condition
- 20 generated tokens per run
- Bootstrap confidence intervals (95% CI, n=1000)

### 4.3 Perturbation Classes

- **Temperature sweeps:**  $T \in [0.1, 0.3, 0.5, 0.7, 0.9, 1.2]$
- **Block resonance:** Block lengths  $n = 1$  to 10 ( $n \times \text{“I”} + n \times \text{“A”}$ )
- **Fourier spectra:** 0.1 to 1.0 Hz, plus pink noise ( $1/f$ ), white noise, chirp

## 5 Results

### 5.1 Final Ranking

Table 3: Complete ranking by flip rate at  $n = 4$ ,  $T = 0.5$

| Model          | Flip Rate | $\sigma_{\max}$ |
|----------------|-----------|-----------------|
| Bloom-560M     | 100%      | -               |
| GPT-2 XL       | 80%       | 0.530           |
| TinyLlama-1.1B | 67%       | -               |
| GPT-2 Small    | 60%       | 0.628           |
| GPT-2 Large    | 53%       | 0.710           |
| Gemma-2-2B     | 17%       | 0.560           |
| GPT-2 Medium   | 7%        | 0.678           |
| DeepSeek-1.3B  | 0%        | 0.609           |

### 5.2 Temperature Sensitivity

Table 4: Temperature sweep results (flip rates at  $n = 4$ )

| Model        | 0.1  | 0.3  | 0.5  | 0.7 | 0.9 | 1.2 |
|--------------|------|------|------|-----|-----|-----|
| Bloom-560M   | 100% | 100% | 100% | 93% | 73% | 10% |
| GPT-2 Small  | 100% | 97%  | 60%  | 30% | 0%  | 3%  |
| GPT-2 Medium | 10%  | 13%  | 7%   | 7%  | 3%  | 7%  |
| GPT-2 Large  | 100% | 93%  | 53%  | 37% | 10% | 0%  |
| GPT-2 XL     | 100% | 97%  | 80%  | 57% | 20% | 0%  |
| TinyLlama    | 100% | 87%  | 67%  | 53% | 33% | 3%  |
| DeepSeek     | 83%  | 17%  | 0%   | 0%  | 0%  | 0%  |
| Gemma        | 0%   | 13%  | 17%  | 7%  | 3%  | 0%  |

### 5.3 Block Resonance

Block length 4 (AAAABBBB) emerges as a universal resonance peak across nearly all architectures.

Table 5: Block resonance peaks

| Model        | Peak Block | Peak Rate | First $\geq 50\%$ |
|--------------|------------|-----------|-------------------|
| Bloom-560M   | 4          | 100%      | 3                 |
| GPT-2 Small  | 5          | 97%       | 3                 |
| GPT-2 Medium | 4          | 100%      | 2                 |
| GPT-2 Large  | 3          | 100%      | 2                 |
| GPT-2 XL     | 4          | 100%      | 2                 |
| TinyLlama    | 4          | 100%      | 2                 |
| DeepSeek     | 4          | 97%       | 4                 |
| Gemma        | 4          | 100%      | 3                 |

## 5.4 Noise Response: Three Dynamical Families

Table 6: Noise response classification

| Model        | Pink Noise (%) | Chirp (%) | Family                   |
|--------------|----------------|-----------|--------------------------|
| GPT-2 Small  | 77%            | 17%       | Broadband Resonator      |
| GPT-2 Medium | 77%            | 17%       | Broadband Resonator      |
| GPT-2 Large  | 77%            | 17%       | Broadband Resonator      |
| GPT-2 XL     | 77%            | 17%       | Broadband Resonator      |
| Bloom-560M   | 13%            | 47%       | Narrowband Resonator     |
| TinyLlama    | 13%            | 47%       | Narrowband Resonator     |
| DeepSeek     | 13%            | 3%        | High-Frequency Selective |
| Gemma        | 10%            | 13%       | Mixed                    |

**Interpretation:** - **Broadband resonators:** Low entrainment threshold; entrained by naturalistic noise - **Narrowband resonators:** Require pure frequency forcing; respond to chirps - **High-frequency selective:** High entrainment threshold; resist entrainment

## 5.5 The Anomalous GPT-2 Medium

GPT-2 Medium (355M) exhibits only 7% flip at  $T = 0.5$ , compared to 53-80% for other GPT-2 models. This suggests a possible critical scale in the GPT-2 family where latent dynamics undergo a qualitative transition.

# 6 Discussion

## 6.1 Latent vs. Output Pathology

A critical distinction:

Table 7: Output pathology vs. latent pathology

| Level            | Example                              |
|------------------|--------------------------------------|
| Output pathology | Repetition loops, degeneration       |
| Latent pathology | Persistent synchronization attractor |

MASSIF measures the latter - precursors that may precede observable output degradation.

## 6.2 AI Safety Implications

Advanced AGI systems may fail dynamically before they fail semantically. Potential dynamical pathologies include: - Recursive collapse - Metastable hallucination states - Self-excitation - Attractor fixation - Unstable planning loops

MASSIF-like systems could function as real-time stability monitors, attractor anomaly detectors, and recursive collapse predictors - the equivalent of cardiovascular monitoring for artificial cognition.

## 6.3 Theoretical Contribution

This work contributes to **Computational Latent Dynamics** - the study of emergent dynamical behavior in high-dimensional learned systems. The results suggest that recursive susceptibility behaves spectrally, not symbolically. Models respond to dynamical structure (periodicity, recurrence, entropy spectra, synchronization pressure) rather than semantic content alone.

## 7 Limitations and Falsifiability

This work establishes empirical phenomenology rather than a complete mechanistic theory. Open questions include:

- Whether persistence flips directly predict downstream generation collapse across all architectures
- Whether resonance bands remain stable at larger scales ( $> 10\text{B}$  parameters)
- Whether susceptibility spectra are tokenizer-dependent
- Whether latent synchronization emerges in multimodal or reinforcement-trained systems
- Whether the observed resonances correspond to specific architectural frequencies, attention periodicities, or dataset statistics

**Falsifiable predictions:** - Block length 4 should produce peak entrainment in any sufficiently large autoregressive transformer - Broadband resonators (GPT-2 family) should show high pink noise sensitivity regardless of scale - Narrowband resonators should require pure frequency forcing for entrainment

Future work should investigate: continuous-time latent dynamics, Lyapunov exponents, attractor dimensionality, cross-layer phase coupling, real-time stability interventions, and latent control systems for AGI safety.

## 8 Conclusion

We introduced MASSIF, a telemetry framework for measuring recursive entrainment in autoregressive language models. Across 8 architectures, we identified three distinct dynamical families based on noise response. GPT-2 Medium is anomalous within its family, suggesting a possible critical scale near 355M parameters. Block length 4 emerges as a universal resonance peak.

We propose **recursive entrainment** as the unifying concept: persistent latent directional coherence induced by phase-alignment between hidden-state trajectories and periodic or self-referential prompt structure.

The results suggest that recursive susceptibility behaves spectrally - models respond to dynamical structure rather than semantic content alone. This work contributes to **Computational Latent Dynamics** and has implications for AI safety, long-context stability, and the design of robust generative systems.

## Data and Code Availability

All code is available at <https://github.com/Ergo-sum-AGI/MASSIF>. Raw data and checkpoints are available on Zenodo (10.5281/zenodo.20208319, 10.5281/zenodo.20232960).

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